CLASS PROJECT RESEARCH REPORT

Trend Following and Mean Reversion ETF Portfolios

Validation and frozen selection

Study endpoint 15 October 2026 · Report prepared 19 October 2026
Scored period 2021-01-01 to 2023-12-31

Executive summary

We compare four unchanged allocations across six fixed long/cash ETF strategies. At baseline costs, the historical daily-cap allocation has cash-excess Sharpe **0.381**, CAGR **6.90%** and daily maximum drawdown **-17.52%**. Cost-matched passive QQQ has Sharpe 0.387, CAGR 9.84% and drawdown -35.12%.

> **Retrospective evidence only.** Configurations, ETF pruning and portfolio weights were chosen using the already examined 2024–2026 sample. Applying those weights backward does not create independent portfolio development or validation. The historical OOS label describes the parent strategy window, not an untouched portfolio test. This distinction matters after extensive candidate and allocation search [4].

> **Historical cap failure preserved.** UNG exceeds one-third of current portfolio capital on 41 daily valuations in each cost scenario; its baseline peak is 35.1826%. We do not refit weights, rebalance or remove these observations.

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1 Project goal

Our objective is to study whether combining causal trend following and oversold-recovery strategies can improve the return and drawdown profile relative to passive ETF exposure, after explicit execution costs. This is a reproducible historical research project, not a trading recommendation.

We ask how the same four static initial allocations behave in different periods, how costs change the result, how much return remains after passive-ETF attribution, and whether the historical capital-share constraint survives backward comparison.

2 Research outline

Trend persistence motivates the trend hypothesis [1], but our ETF implementation is not a replication of that paper. The parent study evaluated 379 definitions per ETF: 289 trend-following and 90 long/cash mean-reversion configurations. Independent development and validation qualification used a strict joint gate at 25-bp slippage: cash-excess Sharpe above matched own-ETF buy-and-hold and CAGR above two-thirds of its CAGR, except that a negative benchmark CAGR requires positive candidate CAGR. Nonfinite candidates cannot pass. Later ETF filtering and correlation pruning led to these six sleeves.

We retain four comparisons: a historical daily-cap Sharpe fit, an initial-only capped fit, an uncapped Sharpe fit and equal initial capital. We exclude the standalone maximum-CAGR 100%-GDX strategy allocation; GDX remains an underlying sleeve. These allocations are saved historical fits, not newly optimized solutions or global-optimum certificates.

3 Data

The parent universe spans 56 US-listed equity-index, international, sector, industry, bond and commodity ETFs. This portfolio report uses GDX, FXI, QQQ, UNG, FDN and BND. Data came from Alpha Vantage daily adjusted records and observed 30-minute bars: native raw captures where available, and matching-vintage-restored legacy captures otherwise. Daily anchors and available 15-minute observations provide same-provider reconciliation checks, not independent-vendor confirmation. The study does not establish point-in-time vendor availability. Daily feature history begins in 2015 where available.

Public inputs here are only saved aggregate statistics, definitions, exact initial weights and audited raster charts. Raw captures, trade-level accounts and full numerical daily paths remain private. Aggregate tables alone cannot reproduce original historical trades.

Scored window	Daily observations	ACT365 years	Initial account
2021-01-01 to 2023-12-31	753	2.984417808219	$1,000

Fresh account opens 2021-01-04T14:30:00+00:00. First daily valuation 2021-01-04T22:00:00+00:00; final valuation 2023-12-29T22:00:00+00:00. These are timestamped accounting observations, not claims of exchange fills at daily valuation time.

4 Backtesting

Indicators are causal and daily, observed with native half-hour checks. Entry and exit require the shared three-trading-hour confirmation, equivalent to seven consecutive half-hour checks, followed by execution at the next eligible observed bar open. Each stage starts flat with $1,000; only feature history is carried, not holdings, orders or confirmation streaks. Accounts are unlevered, fractional long/cash, with the parent study's corporate-action, dividend/reinvestment and paid terminal-liquidation conventions preserved.

Cash accrues at 3% nominal ACT/365, event-compounded. It is neither downloaded Treasury return nor a 3% APY assumption. We do not splice stages into a single wealth curve or manufacture combined trade counts from six independent accounts.

$$V_{c,t} = \Sigma_i \, w_i E_{c,i,t} \quad \text{and} \quad r_{c,t} = [V_{c,t} / V_{c,t-1}] - 1$$

Here c identifies the cost scenario; $E_{c,i,0} = 1$, $V_{c,0} = 1$ and $\Sigma_i w_i = 1$. Each $E_{c,i,t}$ is saved sleeve wealth normalized to its initial account. Static initial wealth mixing is not constant-weight daily return averaging. The same exact weights apply to every cost and period, with no cross-sleeve cash funding or rebalance trades.

Metric definitions

Cash-excess Sharpe is √252 times the mean daily return minus matched cash return, divided by its sample standard deviation with one degree-of-freedom correction. It does not mean outperformance relative to buy-and-hold [2]. Square-root-252 annualization is descriptive and does not supply dependence-robust statistical inference [3]. CAGR uses final normalized wealth and the actual initial-to-terminal ACT/365 clock. Volatility is the annualized sample standard deviation of net daily returns. Drawdown uses daily valuations and includes the initial balance; it is not the worst intraday loss. Matched cash has undefined Sharpe because its cash-excess variance is zero.

Attribution definitions

We fit saved net cash-excess portfolio returns on passive QQQ cash-excess returns with an intercept, and separately on all six cost-matched passive ETF cash-excess returns jointly. The factors are passive buy-and-hold accounts, not active strategy returns. Annual arithmetic alpha is 252 times the daily intercept, not CAGR. Betas are conditional linear sensitivities, not portfolio weights; negative beta does not imply an executed short. R^2 is descriptive, with no significance or causal-skill claim.

5 Execution and slippage

Every purchase and sale pays 1-bp commission plus one-way slippage of 1, 3, 10 or 25 bps. The baseline is 3-bp slippage, or 4 bps total per side. No crypto-course fee assumptions are transferred to ETFs. Each scenario has its own cost-matched passive QQQ and cash control and retains the original paid liquidation. We do not change signals or weights when cost assumptions change.

1 bp slippage each way

Allocation or control	Sharpe	Total return	CAGR	Volatility	Daily max DD	Final $1,000
Daily cap fit	0.391	22.45%	7.02%	11.30%	-17.50%	$1,224.48
Initial cap fit	0.429	25.48%	7.90%	12.55%	-19.22%	$1,254.76
Uncapped fit	-0.231	5.70%	1.88%	4.49%	-9.66%	$1,057.04
Equal initial	0.485	23.81%	7.42%	9.50%	-13.45%	$1,238.13
QQQ buy and hold	0.388	32.39%	9.86%	23.77%	-35.12%	$1,323.90
Matched cash	Undefined	9.37%	3.05%	0.12%	0.00%	$1,093.66

3 bp slippage each way

Allocation or control	Sharpe	Total return	CAGR	Volatility	Daily max DD	Final $1,000
Daily cap fit	0.381	22.04%	6.90%	11.30%	-17.52%	$1,220.37
Initial cap fit	0.419	25.01%	7.77%	12.55%	-19.24%	$1,250.15
Uncapped fit	-0.243	5.54%	1.82%	4.49%	-9.70%	$1,055.36
Equal initial	0.473	23.41%	7.30%	9.50%	-13.48%	$1,234.05
QQQ buy and hold	0.387	32.34%	9.84%	23.77%	-35.12%	$1,323.37
Matched cash	Undefined	9.37%	3.05%	0.12%	0.00%	$1,093.66

10 bp slippage each way

Allocation or control	Sharpe	Total return	CAGR	Volatility	Daily max DD	Final $1,000
Daily cap fit	0.346	20.61%	6.48%	11.30%	-17.59%	$1,206.14
Initial cap fit	0.385	23.42%	7.30%	12.56%	-19.33%	$1,234.16
Uncapped fit	-0.285	4.95%	1.63%	4.48%	-9.85%	$1,049.50
Equal initial	0.432	21.99%	6.89%	9.50%	-13.56%	$1,219.91
QQQ buy and hold	0.385	32.15%	9.79%	23.78%	-35.12%	$1,321.52
Matched cash	Undefined	9.37%	3.05%	0.12%	0.00%	$1,093.66

25 bp slippage each way

Allocation or control	Sharpe	Total return	CAGR	Volatility	Daily max DD	Final $1,000
Daily cap fit	0.272	17.63%	5.59%	11.31%	-17.73%	$1,176.31
Initial cap fit	0.311	20.07%	6.32%	12.58%	-19.52%	$1,200.69
Uncapped fit	-0.375	3.71%	1.23%	4.47%	-10.16%	$1,037.09
Equal initial	0.346	19.03%	6.01%	9.50%	-13.75%	$1,190.25
QQQ buy and hold	0.381	31.76%	9.68%	23.78%	-35.12%	$1,317.56
Matched cash	Undefined	9.37%	3.05%	0.12%	0.00%	$1,093.66

All rows include the additional 1-bp commission each way. “Undefined” is retained, not reported as zero. Full-precision aggregate values accompany this report in CSV and JSON.

6 Weighting

Capital is assigned once to six strategy sleeves, including their idle cash. In particular, the large BND allocation in the uncapped fit is not continuous passive bond exposure. The daily-cap fit was constrained on historical OOS daily sleeve capital shares in all four costs; it is not a future, intraday, ETF-holdings or risk-contribution limit. The initial-only constraint can drift after initialization.

Strategy sleeve	Daily cap fit	Initial cap fit	Uncapped fit	Equal initial
GDX	23.572929%	29.051308%	3.476134%	16.666667%
FXI	24.970310%	27.991974%	4.290936%	16.666667%
QQQ	5.626996%	6.304493%	1.086574%	16.666667%
UNG	23.896545%	26.729329%	4.652412%	16.666667%
FDN	8.631741%	9.922896%	1.147042%	16.666667%
BND	13.301479%	0.000000%	85.346902%	16.666667%

Table weights are rounded for readability. Full-precision JSON is authoritative. The initial-only fit retains its BND weight of $9.720788330706766 \times 10^{-18}$; a displayed 0.000000% is not a zeroed input.

Unchanged strategy definitions

ETF	Saved rule	Exact configuration identity
GDX	Mean reversion: Z-score(20), enter ≤ -2, exit ≥ 0	mr_30m__zscore_20__entry_m2__exit_0__shared_hours_3
FXI	Mean reversion: Z-score(50), enter ≤ -2, exit ≥ 0.5	mr_30m__zscore_50__entry_m2__exit_0p5__shared_hours_3
QQQ	Trend: EMA50 entry / EMA200 exit	v5_30m__entry_ema_50__exit_ema_200__entry_hours_3__exit_hours_3
UNG	Mean reversion: Wilder RSI(14), enter ≤ 30, exit ≥ 50	mr_30m__rsi_14__entry_30__exit_50__shared_hours_3
FDN	Trend: EMA50 entry / EMA100 exit	v5_30m__entry_ema_50__exit_ema_100__entry_hours_3__exit_hours_3
BND	Mean reversion: EMA(50) distance, enter ≤ -2%, exit ≥ +1%	mr_30m__ema_distance_50__entry_m0p02__exit_0p01__shared_hours_3

Observed capital share diagnostics

These diagnostics track each sleeve's share of total saved portfolio wealth at every daily valuation. They do not alter initial weights, trigger cross-sleeve trades or impose new costs. All 16 allocation-cost cases remain in the comparison, whether the observed constraint passes or fails.

Slippage bp	Allocation	Largest daily share	Initial cap	Every daily cap
1	Daily cap fit	35.1551%	Pass	Fail
3	Daily cap fit	35.1826%	Pass	Fail
10	Daily cap fit	35.2785%	Pass	Fail
25	Daily cap fit	35.4828%	Pass	Fail
1	Initial cap fit	38.1974%	Pass	Fail
3	Initial cap fit	38.2354%	Pass	Fail
10	Initial cap fit	38.3679%	Pass	Fail
25	Initial cap fit	38.6514%	Pass	Fail
1	Uncapped fit	85.6060%	Fail	Fail
3	Uncapped fit	85.6140%	Fail	Fail
10	Uncapped fit	85.6417%	Fail	Fail
25	Uncapped fit	85.7008%	Fail	Fail
1	Equal initial	25.4671%	Pass	Pass
3	Equal initial	25.4807%	Pass	Pass
10	Equal initial	25.5283%	Pass	Pass
25	Equal initial	25.6295%	Pass	Pass

> The current daily-cap historical fit fails the backward validation diagnostic: UNG reaches 35.1826% at baseline and exceeds one-third on 41 valuations at each cost. This recorded breach is intentionally retained. No hidden daily rebalancing is assumed.

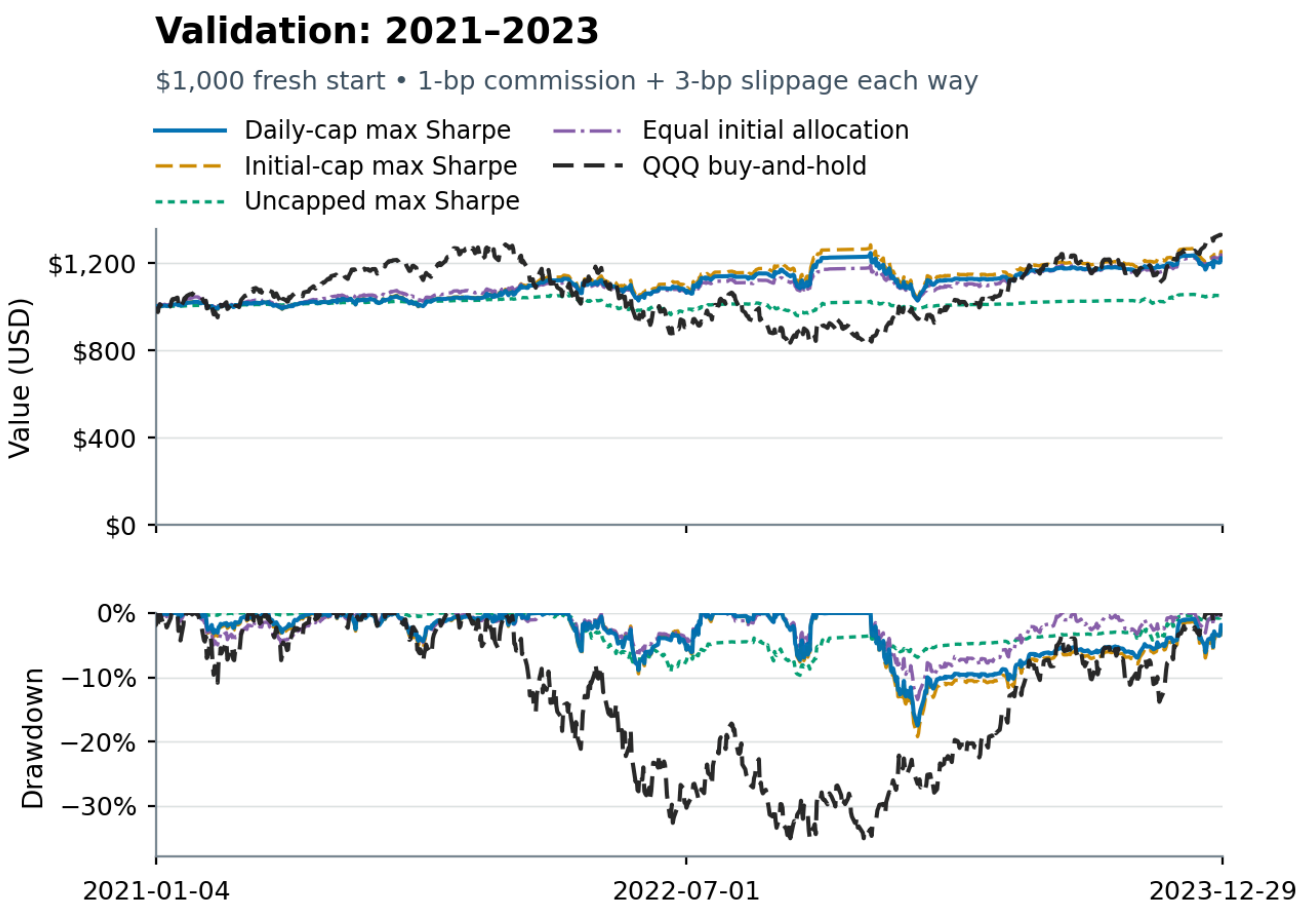
7 Performance evaluation

Period interpretation

Equal initial capital again has the highest baseline Sharpe among the four allocations. The uncapped fit has negative cash-excess Sharpe, while its positive CAGR remains below the matched cash path over the stage. The current allocation has slightly lower Sharpe than passive QQQ, and its historical cap fails; these unfavorable results remain in the report.

For the current allocation, CAGR is 7.02% at 1-bp slippage and 5.59% at 25-bp slippage, with the additional 1-bp commission in both cases. No cost scenario triggers reselection or weight refitting.

Baseline equity and daily drawdown



Static initial allocation; no cross-sleeve rebalancing. Daily peaks include initial capital.

Retrospective fitted allocations; not untouched portfolio OOS. Panels use stage-specific y scales.

Saved-path raster only. Four portfolios and cost-matched passive QQQ each start at $1,000, use 1-bp commission plus 3-bp slippage each way, and include initial balance and paid terminal liquidation. Numerical daily paths are not embedded in this report.

QQQ single factor attribution

The table uses same-stage, same-cost passive QQQ. Alpha is annualized arithmetic percent per year. Betas and R^2 are descriptive fitted statistics; no unreported significance threshold is implied.

Slippage bp	Allocation	Alpha annual	QQQ beta	R squared	Residual vol
1	Daily cap fit	2.76%	0.1796	14.28%	10.46%
1	Initial cap fit	3.56%	0.1975	14.01%	11.64%
1	Uncapped fit	-1.63%	0.0644	11.59%	4.23%
1	Equal initial	2.66%	0.2113	27.95%	8.06%
3	Daily cap fit	2.65%	0.1795	14.26%	10.46%
3	Initial cap fit	3.44%	0.1975	13.99%	11.64%
3	Uncapped fit	-1.69%	0.0644	11.59%	4.23%
3	Equal initial	2.55%	0.2112	27.94%	8.06%
10	Daily cap fit	2.27%	0.1792	14.22%	10.47%
10	Initial cap fit	3.02%	0.1972	13.94%	11.65%
10	Uncapped fit	-1.87%	0.0643	11.60%	4.22%
10	Equal initial	2.17%	0.2109	27.88%	8.07%
25	Daily cap fit	1.46%	0.1786	14.11%	10.48%
25	Initial cap fit	2.13%	0.1967	13.82%	11.68%
25	Uncapped fit	-2.26%	0.0641	11.61%	4.21%
25	Equal initial	1.38%	0.2103	27.75%	8.07%

Joint six passive factor attribution

Each row regresses on passive GDX, FXI, QQQ, UNG, FDN and BND jointly with an intercept. Partial slopes are not separate one-factor betas. Correlated factors can change coefficient interpretation, and higher in-sample R² does not establish predictive value.

1 bp slippage plus 1 bp commission each way

Allocation	GDX β	FXI β	QQQ β	UNG β	FDN β	BND β	Alpha	R²
Daily cap fit	0.0492	0.1138	0.0207	0.0590	0.0376	0.0595	7.03%	38.75%
Initial cap fit	0.0616	0.1241	0.0235	0.0653	0.0422	0.0069	7.89%	38.46%
Uncapped fit	0.0038	0.0274	-0.0017	0.0123	0.0151	0.3620	1.81%	45.89%
Equal initial	0.0322	0.0774	0.0752	0.0419	0.0470	0.0753	6.10%	45.20%

3 bp slippage plus 1 bp commission each way

Allocation	GDX β	FXI β	QQQ β	UNG β	FDN β	BND β	Alpha	R²
Daily cap fit	0.0492	0.1138	0.0209	0.0591	0.0373	0.0594	6.92%	38.74%
Initial cap fit	0.0616	0.1241	0.0237	0.0654	0.0419	0.0067	7.77%	38.46%
Uncapped fit	0.0039	0.0274	-0.0017	0.0123	0.0151	0.3619	1.76%	45.93%
Equal initial	0.0321	0.0774	0.0753	0.0419	0.0468	0.0752	5.99%	45.19%

10 bp slippage plus 1 bp commission each way

Allocation	GDX β	FXI β	QQQ β	UNG β	FDN β	BND β	Alpha	R²
Daily cap fit	0.0491	0.1138	0.0215	0.0592	0.0366	0.0588	6.53%	38.73%
Initial cap fit	0.0615	0.1242	0.0244	0.0656	0.0411	0.0059	7.34%	38.44%
Uncapped fit	0.0039	0.0274	-0.0015	0.0123	0.0149	0.3617	1.59%	46.04%
Equal initial	0.0321	0.0774	0.0758	0.0420	0.0462	0.0748	5.61%	45.15%

25 bp slippage plus 1 bp commission each way

Allocation	GDX β	FXI β	QQQ β	UNG β	FDN β	BND β	Alpha	R²
Daily cap fit	0.0490	0.1139	0.0228	0.0596	0.0349	0.0575	5.71%	38.68%
Initial cap fit	0.0614	0.1244	0.0258	0.0660	0.0393	0.0041	6.43%	38.39%
Uncapped fit	0.0040	0.0273	-0.0013	0.0124	0.0144	0.3610	1.22%	46.17%
Equal initial	0.0319	0.0775	0.0768	0.0422	0.0449	0.0741	4.81%	45.06%

Passive factor overlap diagnostics

Slippage bp	Design rank	Condition number	QQQ VIF	FDN VIF
1	7	5.807	6.752	6.950
3	7	5.807	6.752	6.950
10	7	5.807	6.753	6.950
25	7	5.809	6.754	6.951

Rank includes the intercept. The condition number uses standardized passive factors. QQQ and FDN factor overlap can inflate coefficient uncertainty; a full-rank fit does not eliminate collinearity risk. No p-values or significance claims are reported.

Conclusions and limitations

The four allocations expose a genuine research trade-off: a high fitted Sharpe need not maximize compound growth, and apparent diversification can partly reflect cash-heavy sleeves rather than independent alpha. Comparisons across periods and costs are descriptive robustness checks. We do not select a new winner, alter parent-stage membership, refit weights, or claim the examined OOS sample became untouched again.

The assignment's momentum and reversal themes are addressed through indicator-window trend and oversold-recovery rules, cost sensitivity, allocation comparisons and passive-factor attribution. News/activity filters, seasonal effects, mechanical rebalancing events, liquidity-driven reversals, pairs/spreads and macro-regime-conditioned signals were not separately tested. Those are future extensions, not findings of this report. This long/cash study is not market-neutral statistical arbitrage.

A prospective next step would require a new preregistered release, frozen identities and weights, explicit future evaluation dates, and a practical costed policy if capital-share limits are to be enforced. Nothing in this report authorizes live trading.

Audit and reproducibility

The report is generated only from hash-bound saved aggregate inputs. Companion CSV/JSON retain full precision for all 16 portfolio, 8 control, 16 cap, 16 single-factor and 16 joint-factor rows in this stage. The three reports do not pool returns or substitute another period's clock. Public verification checks source/output hashes, identities, costs, stage boundaries, exact saved weights and chart endpoints; reproducing original trades still requires the private parent-study inputs and engine.

The earlier study uses development 2017–2020, validation 2021–2023 and historical OOS 2024–2026-10-15. A separately preserved corrected-split experiment is inactive and is neither merged into these reports nor used for selection.

Sources and project requirements

The user-provided ClassProject.docx assignment brief supplies the seven-section research outline and required performance dimensions. It is a requirements reference, not a historical result source or authorization for new tests. Numerical sources are the repository's four_portfolios_three_periods aggregates, four saved portfolio definitions, six strategy definitions, all-period attribution tables and audited baseline PNGs. See each stage manifest and the root report manifest for exact portable input/output SHA256 bindings.

Methodology references

[1] Moskowitz, T. J., Ooi, Y. H., and Pedersen, L. H. (2012). Time Series Momentum. Journal of Financial Economics 104, 228–250. Primary research page: https://www.aqr.com/Insights/Research/Journal-Article/Time-Series-Momentum

[2] Sharpe, W. F. (1994). The Sharpe Ratio. Journal of Portfolio Management. Author text: https://web.stanford.edu/~wfsharpe/art/sr/sr.htm

[3] Lo, A. W. (2002). The Statistics of Sharpe Ratios. Financial Analysts Journal 58, 36–52. Author research page: https://alo.mit.edu/research-page/the-statistics-of-sharpe-ratios/

[4] Bailey, D. H., Borwein, J., López de Prado, M., and Zhu, Q. J. (2017). The Probability of Backtest Overfitting. Journal of Computational Finance. DOI 10.21314/JCF.2016.322. University repository: https://scholarworks.wmich.edu/math_pubs/42/

References motivate methodology and caution, not the numerical outcomes. We do not claim a paper replication, a computed PBO or CSCV estimate, or inferential confidence intervals.